

Jawaharlal Nehru Govt. Engineering College Sundernagar, Mandi (H.P.)
Course Name: Applied Chemistry Course Code: CHM-111 Examination: 1st Periodical
Scheme: CBCS Branch: CE/CSE Semester: 2nd
Question Paper for 1st Periodical Examinations (6th March, 2023)

Time: Two hours

Max. Marks: 50

Note: All questions are compulsory.

Course Outcomes: *At the end of course students will be able to:*

1. Apply the concept of water treatment and electrochemistry in daily life and industrial applications.
2. Design various types of methods to prevent corrosion.
3. Apply the concept and principles of spectroscopic techniques in material characterization.
4. Evaluate various types of fuels on the basis of their properties and uses.
5. Summarize role of chemistry in ICT.
6. Critically examine the role of various nano-materials and polymers in engineering field.

Part A

Q1.

- (i) (CO1) Differentiate between BOD and COD?
- (ii) (CO1) Calculate the temporary and permanent hardness in a pre-boiled water sample containing $\text{Mg}(\text{HCO}_3)_2 = 14.6 \text{ mg/L}$, $\text{MgCl}_2 = 9.5 \text{ mg/L}$, $\text{CaSO}_4 = 6.8 \text{ mg/L}$.
- (iii) (CO2) What are corrosion inhibitors? Give suitable examples.
- (iv) (CO4) How petroleum reforming improves the quality of fuel?
- (v) (CO6) List out the monomers present in PET and Starch.

(2×5=10)

Part B

- Q2. (CO1) (i) Discuss in detail the processes involved in domestic water purifiers. **(5)**
(CO1) (ii) Write short notes on water quality parameters; hardness and turbidity. **(5)**
- Q3. (CO2) Discuss in detail about the various corrosion preventive methods. **(10)**
- Q4. (CO6) (i) Calculate the net and gross calorific value of a coal sample having the following composition: C=80%, H=5%, O=4%, N=3%, S=3.5% and ash=5%. **(5)**
(CO6) (ii) How composition of nitrogen is determined using ultimate analysis of coal. **(5)**
- Q5. (CO6) Give synthesis, properties and applications of following polymers: PMMA, Lexan and Kevlar. **(3,3,4=10)**

Course Advisor: Dr. Ajnesh Singh